Sveconorwegian palaeomagnetism in hyperite dolerites and syenites from Scania, Sweden

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Proterozoic hyperite dolerites and syenites from the Scanian part of the Protogine Zone (Sveconorwegian Front) have the same characteristic remanence direction after alternating field demagnetization. The direction is $D = 297.2°$, $I = -83.8°$, $\alpha_{95} = 3.7°$, $k = 114.6$, which gives a palaeopole at 49°S, 211°E, $d_p = 7.1°$, $d_m = 8.3°$. The pole position, and a majority of K–Ar datings on the rocks studied, implies that they became remagnetized during the Sveconorwegian regeneration. This new remanence was not affected by a later intrusion of a dolerite swarm in the intersecting Fennoscandian Border zone. A tentative apparent polar wander path for the Sveconorwegian is presented. □ *Palaeomagnetism, thermomagnetism, remagnetization, demagnetization, pole positions, polar wandering, intrusions, hyperite dolerite, syenite, Sveconorwegian, Proterozoic, Protogine Zone, Tornquist Lineament, Fennoscandian Border Zone, Skåne, southern Sweden, N5533 N5650 E1484 E1250.*

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Two important tectonic lineaments intersect in Scania (Skåne), southernmost Sweden. One is the zone separating the Precambrian Fennoscandian Shield from central Europe and its sedimentary cover and the other is the Protogine Zone (PZ) which constitutes the border between the entirely different Proterozoic terrains of the southwestern and central Fennoscandian Shield. Movements and intrusions in these zones have contributed to the mosaic pattern and the mix of Proterozoic gneisses, granites and Phanerozoic sediments that form the geology of Scania. The purpose of this paper is to present a study of the magnetic properties of intrusive hyperite dolerites and syenites from the Scanian part of the PZ in relation to their situation in the intersection between those tectonic lineaments.

The Protogine Zone – other names sometimes used for this feature are the Schistosity Zone, The Gothide Front, the Sveconorwegian Front – trends from Scania northwards across southern Sweden to Norway where it is cut by the Scandinavian Caledonides (Fig. 1). It separates the Sveconorwegian southwestern corner of the Fennsocandian Shield from its central Svecokarelian part. There is at present a running discussion whether the PZ is part of a global suture, the scar of a plate collision between a "Grenvillian–Sveconorwegian" plate and the Laurentian–Fennoscandian plate or whether it is an intrashield structure along which movements have occurred intermittently during the Precambrian, i.e. Donaldson et al. (1973), Zeck & Malling (1976), Patchett & Bylund (1977), Piper (1980), Gorbatschev (1980), and Lindh et al. (in press).

The PZ is approximately 15–30 km wide in its southern part, the borders being undetermined due to the lack of outcorps. The rocks within the southern part of the Zone are various granites and gneisses and intrusive in those occurs a set of hyperite dolerites and syenites, the subject of this study. The Zone is covered by younger sediments in the southern half of Scania, where it only outcrops on the horst Romeleåsen (Fig. 1).

The Sveconorwegian block west of the PZ present Rb–Sr ages from 1700 to 900 Ma while the K–Ar ages concentrate around 1000 to 800 Ma (Welin & Blomqvist 1966). This is in contrast to the Svecokarelian block east of the Zone where there is a general agreement between Rb–Sr and K–Ar ages with only local resetting of the later (Åberg 1978). It has been suggested that the Sveconorwegian differences in radiometric ages was due to a regional heating and regeneration of the areas, followed by uplift and accompanied by a re-opening and later closing of the K–Ar system when the blocking temperature was reached during the later stages of uplift

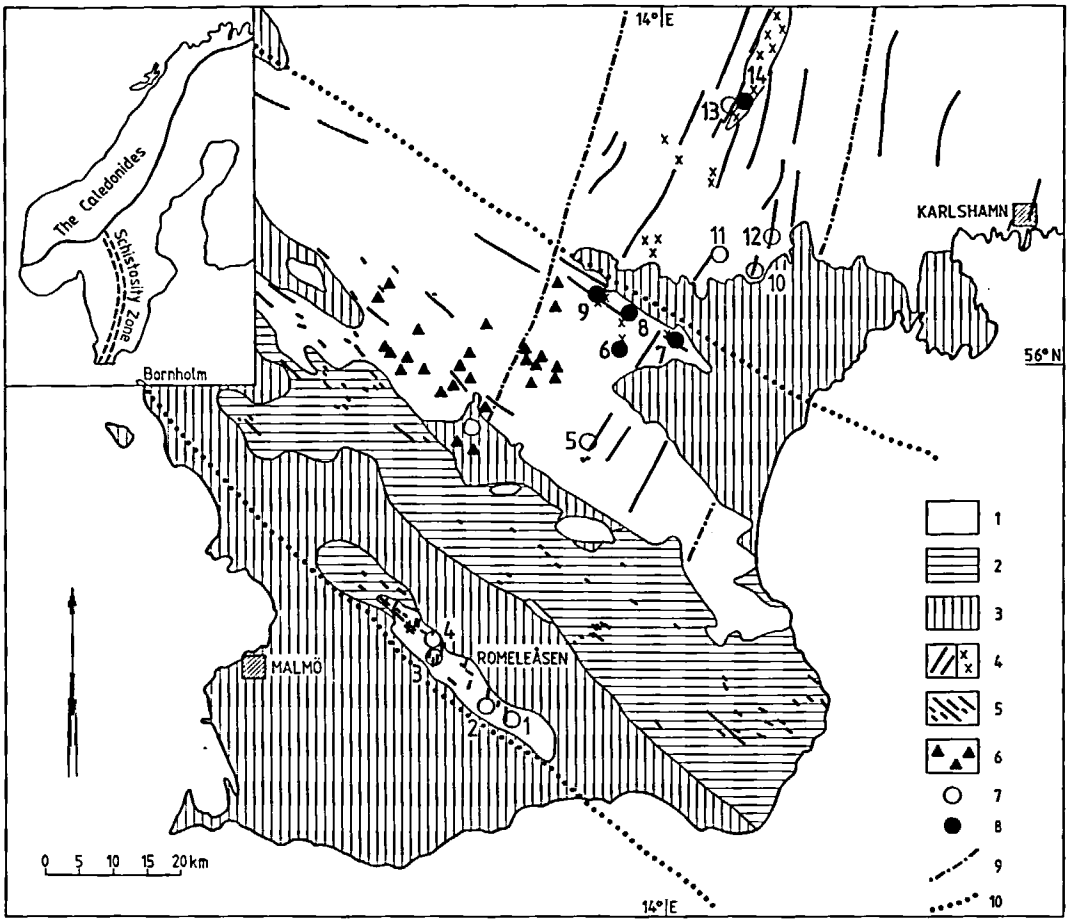


Fig. 1. Geological map of Scania with sampling sites. (Simplified after Stanfors et al. 1966.) Syenite occurrences from the geological map-sheets Kristianstad SO and NO. 1 = Proterozoic basement; 2 = Palaeozoic sediments; 3 = Mesozoic sediments; 4 = Proterozoic dykes (left) and syenites (right); 5 = NW-trending Palaeozoic dykes; 6 = Mesozoic basalts; 7 = sampled hyperite dolerite; 8 = sampled syenite; 9 = border of Protogine Zone; 10 = margin of Fennoscandian Border Zone.

(Welin & Blomqvist 1966). This theory was emphasized in a radiometric and palaeomagnetic study by Patchett & Bylund (1977) on dolerites situated east of the PZ and orientated parallel with it. It was proposed that the dolerites intruded in tension cracks formed on the western edge of the Svecokarelian block due to stresses caused by uplift of the western block.

Most of Scania is situated within the Fennoscandian Border Zone, which together with the Danish-Polish Trough forms the Scandinavian part of the Tornquist Lineament. In Scania it is outlined in a set of horsts of Proterozoic gneisses and granites and a large swarm of NW-trending dolerite dykes.

While movements in the PZ occurred mainly in the Proterozoic, the Tornquist Lineament has been active during the Phanerozoic with a possible peak in the late Palaeozoic (Hjelmqvist 1939). However, movements and intrusions have occurred later (Bölau 1972; Hjelmqvist 1975), one evidence being the cluster of volcanic plugs, the remains of an extensive alkaline volcanism in central Scania during the Jurassic–Cretaceous (Norin 1934; Klingspor 1976).

Palaeomagnetic studies have previously been made on the NW-trending dolerites (Mulder 1971; Bylund 1973, 1974) and on the alkaline basalts (Bylund 1973; Schweitzer 1975). Mulder (1971) presented results from five hyperite dolerite dykes from northern Scania and further north in the PZ. Claesson (1977) made a palaeomagnetic study of the Scanian Palaeozoic sediments.

Geological setting and age relations

The hyperite dolerites in Scania generally trend NNE, parallel with the PZ. They vary in width from a few metres up to 200–300 m. There is a range from unaltered dykes with chilled margins to dykes completely amphibolitized. The ratio of alteration seems to be related to the degree of schistosity of the country rock. Kornfält et al. (1978) note that the deformation of the rocks within the Zone is restricted to narrow bands with unaffected rocks in between, and that the degree of alteration depends on the location of the dykes in relation to those deformed zones. Structures in the Zone imply that vertical movements have prevailed. The dykes stand vertical or subvertical.

The hyperite dolerites contain both clino- and orthopyroxene and due to the presence of the latter in the form of hypersthene, the terms *hyperite* and *hyperite dolerite* have been used for this and similar rocks connected with the PZ. The plagioclase, An 35–48%, in the hyperite dolerites is coloured by a pigment that gives the rock a black to brown appearance. Olivine occurs in some of the dykes studied. Opaques, mainly as titanomagnetite, exists in amounts between 6 and 15 wt% (Hjelmqvist 1934; Kornfält et al. 1978). For an account of the petrology and chemistry of the hyperite dolerites, see Hjelmqvist (1934), Kornfält et al. (1978), and Lindh et al. (in press).

The syenites occur in massifs of varying size, from 1 to 150 km^2, in the northern half of the area studied (Fig. 1). Similar rocks, the Vaggeryd syenites, are found 160 km further north in the PZ (Quensel 1960). The scanian syenites are medium- to coarse-grained rocks ranging from quartz syenites to syenites and monzonites. Main minerals are potassium feldspar, plagioclase, hornblende, and quartz, with minor amounts of biotite, pyroxene, zircon, apatite, and calcite. Opaques constitute 1–2%. Schistose parts both whitin and on the margins of the syenite massifs reflect their involvement in the tectonic movements in the PZ.

Rb–Sr whole-rock dating give ages of 1565 ± 230 Ma for the hyperite dolerites and 1184 ± 38 for the syenites (Klingspor 1976). (All radiometric ages cited have been recalculated using the decay constants recommended by Steiger & Jäger 1977.) The presence of hyperite dolerite intrusions in granites of the Karlshamn type indicates a somewhat younger age for the hyperite dolerites. The Karlshamn granite has been dated to 1422 ± 31 Ma (Rb–Sr whole-rock, Sprinnger 1979). Mulder (1971) presented K–Ar ages between 786 and 1569 Ma from similar hyperite dolerites further north in the PZ. Klingspor (1976) gave a similar range, 679 to 1595 Ma for the hyperite dolerites and 880 to 1185 for the syenites in Scania. She presented histograms, which show that of 37 K–Ar datings of the hyperite dolerites 19 fall between 750 and 1050 Ma, for the syenites 4 out of 6 are between 750 and 900 Ma. She supported the idea of Welin & Blomqvist (1966) and Mulder (1971) that the concentration of K–Ar ages in the range 800–1000 Ma is due to a Sveconorwegian regeneration and uplift of the western block.

Sampling and measurement

Orientated samples were collected either by hand or with the help of a light core-drill. The orientation was made by sun-compass or magnetic compass and the samples were marked while still attached to the outcrop. From each hand sample or core were obtained one to four cylindrical specimens for measurements. Most of the sampling was done in quarries and road-cuts.

The natural remanent magnetization (NRM) was measured on a 85 Hz JFG-JR3 or a 7 Hz Digico spinner magnetometer. Alternating field (AF) demagnetization was used to remove masking soft magnetism and isolate a characteristic stable magnetic component. Tests were also performed with thermal demagnetizaion. All demagnetization was made in near field-free space, less than 10 nT (nanotesla). An estimate of the Curie temperature was made by measuring the saturation magnetization (J_s) against temperature.

Results

The results are listed in Table 1 and typical examples of specimen behaviour during demagnetization are presented in Figs. 2–3. Sample means were used in the calculation of site directions. Fisher (1953) statistics were used for calculating mean directions. There was no significant difference observed between the hyperite dolerites and the syenites during the tests and the following discussion applies to both rock types.

The progressive AF-demagnetization generally showed an increase in intensity accompanied by directional changes during the first steps followed by a smooth decrease and stable direc-

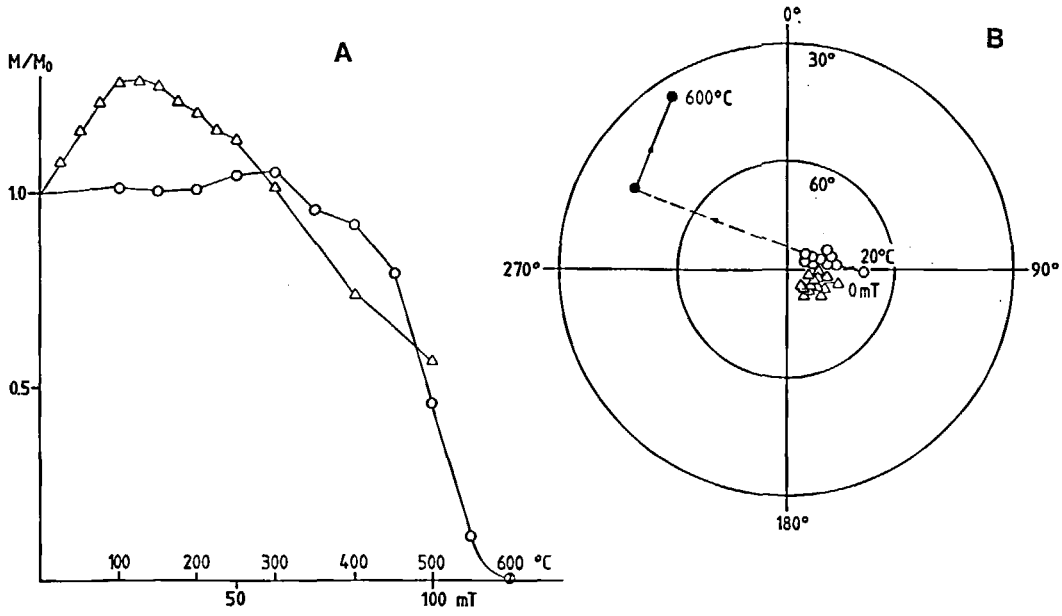


Fig. 2. Behaviour of hyperite dolerite specimens during progressive AF-(triangles) and thermal (circles) demagnetization. □ A. Normalized curves for the magnetic intensity during demagnetization. □ B. Direction changes during demagnetization. Open symbols denote negative inclination (pointing up) and closed positive inclination. Only the central part of the stereogram is used.

tions (Figs. 2–3). This behaviour can be interpreted as due to the presence of two magnetic components of which one has a low coercivity and a steep positive inclination, while the stable component has a characteristic steep negative inclination. The unstable component may be attributed to influences from the Earth's present field.

The thermal demagnetization tests showed similar results (Fig. 2–3). The thermal decay curves (Figs. 2–3) show a rapid decrease in intensity between ca. 500 and 570°C indicating a corresponding blocking-temperature range for the magnetic minerals.

Site mean directions after demagnetization are given in Fig. 4.

The measurements of the saturation magnetization against temperature (Fig. 5) indicated Curie temperatures close to that for magnetite (578°C). The heating and cooling curves are reversible, which indicates that no chemical alterations of magnetic importance occurred during the heating process. It was performed in air.

A microscopic study of the opaque minerals showed titanomagnetite with exsolution lamellae of ilmenite and ilmenite as the main ore minerals together with minor amounts of sulphides. The titanomagnetite belongs to classes I–III in Watkins & Haggerty's (1967) classification of high temperature oxidation of titanomagnetite.

The grain size of the opaques varied, from 1 mm down to close to the resolving power of the microscope (ca. 0.5 μm). The microscopic study, the results from the thermal demagnetization and the J_s/T measurements indicate that titanomagnetite, close to the magnetite end in the Ulvö spinel–magnetite series, carries the NRM in both the hyperite dolerites and the syenites.

The initial NRM in the hyperite dolerites ranges between 1 and 10 Am^{-1}, the susceptibility from 6×10^{-6} to 600×10^{-6} (SI-units) and the Q-values between 0.1 and 2.2. For the syenites the corresponding values were 2×10^{-3}–250×10^{-3} Am^{-1}, 2×10^{-6}–600×10^{-6} and 0.1–1.2, respectively. An account of magnetic susceptibility, Q-values and densities of rocks in the PZ is given in Kornfält et al. (1978).

Discussion

Aeromagnetic measurements on the geological map-sheet Kristianstad SO (Kornfält et al. 1978) show a large amount of previously unknown NW-trending dykes. This demonstrates that the

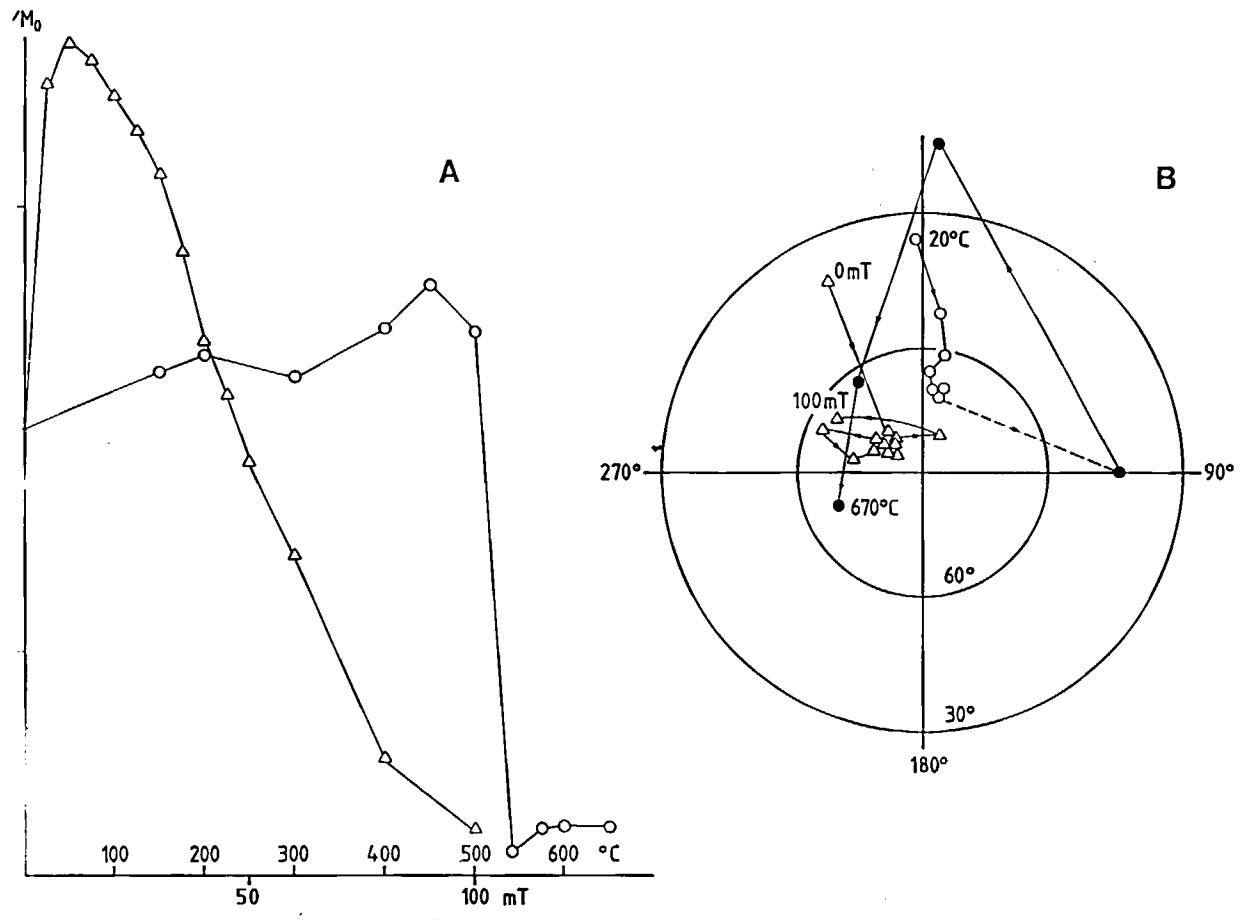


Fig. 3. Behaviour of syenite specimens during progressive demagnetization. Conventions as in Fig. 2.

amount of quartz dolerite dykes that intruded in the Fennoscandian Border Zone in the Late Carboniferous–Early Permian is much larger than hitherto considered. It could be expected that the intrusion of such an extensive dyke swarm would have heated the rocks of the area on a regional scale and to such an extent that the magnetic properties of the rocks would have been affected. This does not seem to have been the case. There is no difference in magnetic directions from hyperite dolerites sampled in areas with a large amount of younger dykes and those from the north-eastern corner of Scania where they are virtually absent. Further, no magnetic component that could have been attributed to the quartz dolerites was found during the demagnetization tests. Also there is a large difference in palaeomagnetic pole positions between the hyperite–dolerites–syenites and the younger dolerites. (Fig. 6).

The position of the combined hyperite-dolerite and syenite pole together with poles of Sveconorwegian age (Table 2, Fig. 6), implies that their stable magnetism is of that age, in compliance with the majority of the K–Ar datings. This suggests that the regional heating of the block west of the PZ continued into the Zone and reset both the K–Ar ages and the magnetic directions. This process or remagnetization along the ambient field due to heating and a later blocking-in of that magnetic direction during slow cooling and uplift has been demonstrated from Greenland by Beckman & Mitchell (1975), Morgan (1976), and from the Scandinavian Caledonides by Bylund & Zellman (1980). The theoretical considerations of the process have been treated by Pullaiah et al. (1975).

The hyperite-dolerite and syenite pole is situated at the edge of a group of Sveconorwegian poles. This group is in Table 2 and Fig. 6 divided

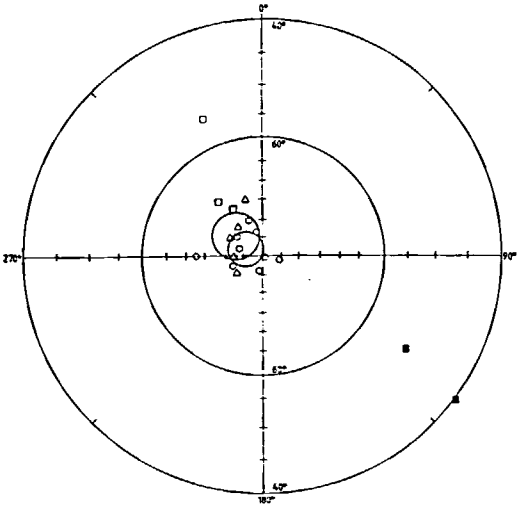


Fig. 4. Site mean directions after AF-demagnetization. ○ = hyperite dolerites; △ = syenites; □ = hyperite dolerites studied by Mulder (1971) and listed in Table 1. The small circle is the 95% confidence circle for the Scanian hyperite dolerites, the larger one is for the syenites. Open symbol denotes negative inclination. Only the central part of the stereogram is used.

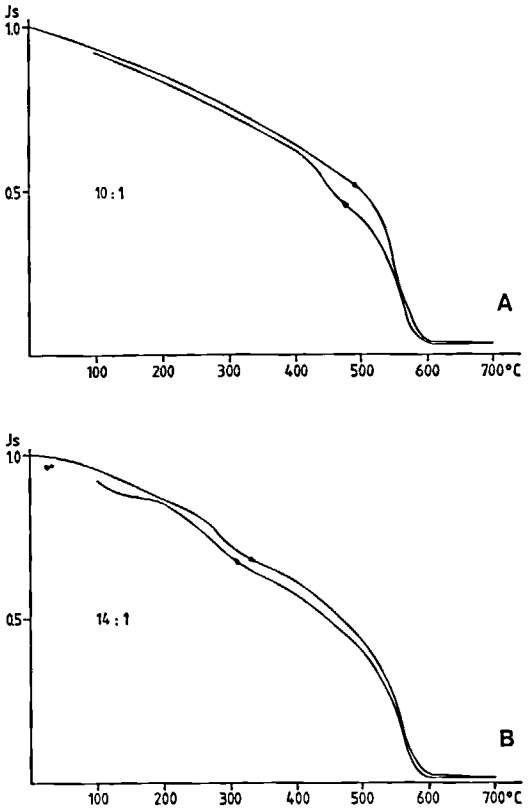


Fig. 5. Saturation magnetization (J_s) *versus* temperature for the hyperite dolerite (A) and syenite (B). The scale on the J_s-axis is arbitrary.

into poles from the Sveconorwegian block proper, from within the PZ, and from dolerites intruded during the uplift into tension cracks east of the PZ.

The poles of the Jotnian dolerites (Table 2, Fig. 6) form the end of the 1800–1200 Ma apparent polar wander path (APWP) for Fennoscandia, and the distance between them and the Sveconorwegian group is ca. 50°. There is also an age gap, from the end of the Jotnian magmatic event at ca. 1200 Ma to the time for the blocking-in of the new magnetism in the Sveconorwegian rocks. This gap is difficult to estimate but may be in the order of 150 to 200 Ma. There are at present no palaeomagnetic poles from Fennoscandia covering that age gap and that presents several possibilities for connecting the Jotnian dolerite poles to the Sveconorwegian poles. Besides, there is as yet no clear correlation between pole position and radiometric age in the Sveconorwegian group. Various models, however, have been proposed for the Fennoscandian APWP during Sveconorwegian time and its connection with the older part, i.e. by Spall (1973), Poorter (1975), Patchett & Bylund (1977), Abrahamsen (1977), and Piper (1980). The currently preferred model is a tight loop similar to the "Grenville loop" proposed for the contemporaneous Grenville belt in the Laurentian Shield

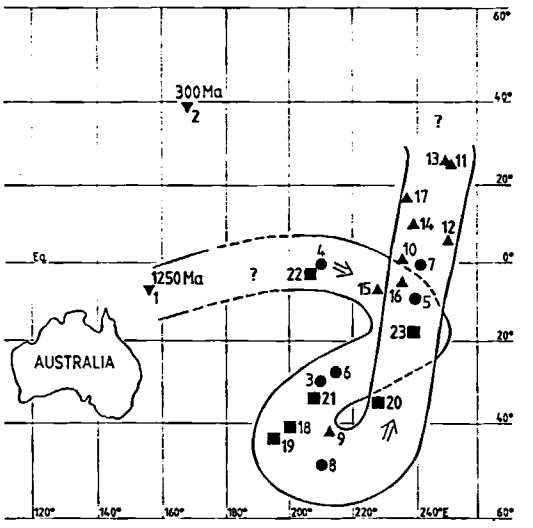


Fig. 6. Sveconorwegian palaeopoles. ▲ = from east of the PZ, ● = within the PZ; ■ = from west of the PZ; ▼ = poles from Jotnian dolerites (1) and NW-striking late Palaeozoic dolerite dykes (2). Figures refer to Table 2.

Table 1. Magnetic directions after AF-demagnetization. N = number of samples. AF = demagnetizing field with least scatter in directions and presented in this table. α_{95} = semi-angle of the 95% cone of confidence. k = the best estimate of Fisher's (1953) precision parameter. R = length of the resultant vector, each sample given unit weight.

Site, rock type	N	AF (*mT*)	Decl. (°)	Incl. (°)	α_{95} (°)	k	R
1. Belinga, Hd	6	20	269.6	–72.4	9.8	48.2	5.8962
2. Stenberget, hd	3	30	198.0	–86.5	–	–	2.9653
3. Romeleklint, hd	6	50	249.9	–81.8	2.8	587	5.9910
4. Krutladan, hd	13	30	285.6	–83.7	6.1	46.5	12.7421
5. Bjärahall, hd	21	50	346.6	–83.8	2.8	132.7	20.8493
6. Djurröd, sy	9	20	299.5	–79.3	7.3	50.6	8.8418
7. Önnestad, sy	5	20	342.6	–74.0	14.5	28.9	4.8616
8. Rosenlund, sy	8	50	239.3	–81.8	6.7	68.7	7.8981
9. Brantaberg, sy	10	20	269.6	–82.1	6.3	60.4	9.8511
10. Balsberget, hd	12	20	146.4	–89.4	5.7	59.2	11.8141
11. Bjärlöv, hd	7	20	306.1	–81.4	5.7	114.6	6.9477
12. Röetved, hd	6	20	107.1	–86.6	8.6	61.6	5.9188
13. Rumperöd, hd	4	50	339.6	–79.7	5.6	269.6	3.9889
14. Ekeröd, sy	11	10	319.9	–79.7	4.7	93.6	10.8931
Mean hyperite dolerites hd (sites)	9	–	290.5	–85.1	4.8	114.4	8.9301
Mean syenites, sy (sites)	5	–	303.3	–81.5	7.0	120.1	4.9667
Mean all sites	14	–	297.2	–83.8	3.7	114.6	13.8866
Magnetic directions for hyperite dolerites in the Protogine Zone presented by Mulder 1971 and discussed in the text							
Hägghult, hd	50	–	327	–75	1	338	
Osby, hd	5	–	337	–53	6	163	
Målaskog, hd	5	–	124	+51	2	1892	
Hjortsjö, hd	5	–	322	–72	2	1372	
Kristinehamn, hd	5	–	128	+39	7	117	

(McWilliams & Dunlop 1978). This model does not need the existence of two colliding plates for explaining the palaeomagnetic data (Patchett & Bylund 1977). In Fig. 6, a "Sveconorwegian loop", modified after Piper (1980) is presented. In it there is a certain concentration of poles in the bends, which may represent periods of large tectonic activity (Irving & Park 1972). As the curve partly overlaps itself it is difficult to assign relative ages to several of the poles, e.g. Nos. 5, 7, 10, 16, and 23. The poles from the Sveconorwegian block and the PZ are distributed along most of the curve with the combined hyperite-dolerite and syenite pole from Scania (8) at the southern bend. Poles 3–7 represent hyperite dolerites from elsewhere in the PZ (Mulder 1971). Of those, poles 3, 4, and 6 are from dykes with steep negative magnetic inclinations and directions similar to those of the Scanian rocks studied, while 5 and 7 have intermediate positive inclination and nearly opposite directions (Table 1, Fig. 4). Mulder interpreted this discrepancy as caused by a field reversal. This may be the case, but if one set is turned 180°, there is a difference of 24° between the two group directions. This can be due to a time-difference in the acquisition of the NRM, during which time apparent polar wander occurred. (Pesonen 1979) or to an intrinsic feature of the geomagnetic field, e.g. an offset dipole (Wilson 1972). The first suggestion implies different periods of remanence acquisition in different parts of the PZ due to variations in the rate of uplift or thermal influence, and points to a possible method of relative dating of events in the Zone.

The dolerites that intruded east of and parallel with the PZ, yield, with the exception of the Karlshamn dyke (9), poles in the upper part of the "Sveconorwegian APWP" (10–17). Those dolerites have Rb–Sr intrusion ages between 1000 and 870 Ma (Patchett 1978) and their pole positions indicate that they represent the last magmatic event in the Sveconorwegian orogeny. Included in those results is the pole (17) from the Listed and Bølshavn B dolerites on the island of Bornholm, SE of Scania (Abrahamsen 1977). The Karlshamn dyke has a steep positive magnetic direction and a corresponding pole position

Table 2. Sveconorwegian poles together with Jotnian and late Palaeozoic dyke poles used for comparison. d_p and d_m are halfaxes in the 95% ellipses of confidence.

Rock unit	Pole position lat.	long	d_p (°)	d_m (°)	Radiometric age (Ma)	Method	Reference (palaeopole and age)
1. Jotnian dolerites, mean	7.2°S	156.9°E	2.0	3.1	1247–1213	Rb–Sr, K–Ar	Piper 1979; Welin & Lundqvist 1975; Patchett 1978
2. NW dolerites, Scania, mean	38.5°N	168.5°E	3.3	6.7	300±4	K–Ar	Bylund 1974; Klingspor 1976
Hyperite dolerites and syenites within the Protogine Zone							
3. Hägghult, hd	30°S	211°E	–	–	843±25	K–Ar	Mulder 1971
4. Osby, hd	2°S	213°E	–	–	786±25		
5. Målaskog, hd	10°S	239°E	–	–	891±45		
6. Hjortsjö, hd	27°S	216°E	–	–	1569±50		
7. Kristinehamn, hd	2°S	241°E	–	–	1513±50		
8. Scania, hd + sy	49.1°S	210.7°E	7.1	8.3	1595–679	Rb–Sr, K–Ar	This paper; Klingspor 1976
Dolerites east of the Protogine Zone							
9. Karlshamn	43.2°S	213.5°E	8.9	9.3	936±56	Rb–Sr	Patchett & Bylund 1977
10. Tärnö	0.3°N	237.0°E	3.6	5.9	871±21		
11. Bräkne-Hoby	22.2°N	251.9°E	3.1	6.1	975±58		
12. Väby	5.0°N	249.6°E	–	–	–	–	
13. Fäjö	23.3°N	248.9°E	–	–	–	–	
14. Nilstorp	9.0°N	238.5°E	5.8	10.6	985±65	Rb–Sr	
15. Årby	7.3°S	227.4°E	6.8	9.5	995±65		
16. Falun	6.1°S	237.6°E	4.6	7.1	966±20		
17. Listed and Bølshavn B (Bornholm)	17.7°N	237.3°E	3.2	6.2	–	–	Abrahamsen 1977
Rocks west of the Protogine Zone							
18. Egersund, anorthosite	42°S	200°E	$\alpha_{95}=15°$		ca. 1000	K–Ar	Hargraves & Fish 1972
19. Farsund, farsundite	43.3°S	194°E	9.3	9.7	920±20	^{207}Pb–^{206}Pb	Murthy & Deutsch 1975
20. Rogaland, basement	36°S	227°E	10	12	950–850	Rb–Sr, K–Ar	Poorter 1972
21. Rogaland, Hunnedalen dykes	34°S	208°E	9	10	Late Proterozoic		
22. Bamble–Kongsberg amphibolite and hyperite	3°S	207°E	6	8	1120–975	K–Ar	Poorter 1975
23. Tuve, Göteborg, dolerite	17°S	239°E	2	3	–	–	Abrahamsen 1974

close to that of the Scanian hyperite dolerites and syenites. This implies that this dyke intruded earlier than the other "eastern" dykes, close in time to the remagnetization of the rocks in the PZ in Scania.

Conclusions

The results of this study imply that the original NRM of the Scanian hyperite dolerites and syenites was overprinted or obliterated due to events related to the Sveconorwegian orogeny, viz. regional heating, uplift, and slow cooling. This later remanence became directed along the ambient geomagnetic field. Later events in the Protogine Zone and the intersecting Tornquist Lineament, e.g. the intrusion of a large swarm of dolerite dykes, did not affect this NRM.

The palaeomagnetic data for the whole Sveconorwegian belt, including the Protogine Zone, show a complicated pattern due to differences in thermal influence, magnetic minerals with varying blocking-temperature spectra, and different rates of cooling and uplift in the Sveconorwegian belt and the Protogine Zone. This makes any APWP based on present data highly tentative. K–Ar and other radiometric datings together with palaeomagnetic studies of rocks with different magnetic properties may solve the problems inherent in the Sveconorwegian polar wander path.

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References

Åberg, G., 1979: Precambrian geochronology of southeastern Sweden. *Geologiska Föreningens i Stockholm Förhandlingar 100* [for 1978], 125–154.

Abrahamsen, N., 1974: The palaeomagnetic age of the WNW.-striking dikes around Gothenburg, Sweden. *Geologiska Föreningens i Stockholm Förhandlingar 96*, 163–170.

Abrahamsen, N., 1977: Palaeomagnetism of 4 dolerite dikes around Listed, Bornholm (Denmark). *Bulletin of the Geological Society of Denmark 26*, 195–215.

Beckman, G. F. J. & Mitchell J. G., 1976: Palaeomagnetic and geochronological work in central west Greenland. *Earth and Planetary Science Letters 30*, 269–280.

Bölau, E., 1972: Genese und Alter der Horste Schonens. *Geologiska Föreningens i Stockholm Förhandlingar 94*, 411–422.

Bylund, G., 1973: *Paleomagnetic study of Scanian dolerites and basalts.* 53 pp. Lunds universitet, Lund. [Unpublished Ph. D. thesis.]

Bylund, G., 1974: Palaeomagnetism of dykes along the southern margin of the Baltic Shield. *Geologiska Föreningens i Stockholm Förhandlingar 96*, 231–235.

Bylund, G. & Zellman, O., 1980: Palaeomagnetism of the dolerites of the Särv Nappe, southern Swedish Caledonides. *Geologiska Föreningens i Stockholm Förhandlingar 102*, 393–402.

Claesson, K. C., 1977: *Palaeomagnetic studies of Swedish Lower Palaeozoic rocks.* 349 pp. School of Physics, The University, Newcastle-upon-Tyne. [Unpublished Ph. D. thesis.]

Donaldson, J. A., McGlynn, J. C., Irving, E. & Park, J. K., 1973: Drift of the Canadian Shield. *In* D. H. Tarling & S. K. Runcorn (eds): *Implications of continental drift to the Earth sciences*, 3–17. Academic Press, London.

Fisher, R., 1953: Dispersion on sphere. *Proceedings of the Royal Society of London A 217*, 295–305.

Gorbatschev, R., 1980: The Precambrian development of southern Sweden. *Geologiska Föreningens i Stockholm Förhandlingar 102*, 129–136.

Hargraves, R. B. & Fish, J. R., 1972: Paleomagnetism of anorthosite in southern Norway and comparison with an equivalent in Quebec. *Geological Society of America, Memoir 132*, 141–148.

Hjelmqvist, S., 1934: Zur Geologie des südschwedischen Grundgebirges. Die kristallinischen Gesteine des Romeleåses. *Meddelanden från Lunds Geologiska-Mineralogiska Institution 58*, 1–183.

Hjelmqvist, S., 1939: Some post-Silurian dykes in Scania and problems suggested by them. *Sveriges geologiska undersökning C 430*, 1–24.

Hjelmqvist, S., 1975: A Rhenish fault on the southern border of the Baltic Shield. *Geologiska Föreningens i Stockholm Förhandlingar 97*, 89–91.

Irving, E. & Park, J. K., 1972: Hairpins and superintervals. *Canadian Journal of Earth Sciences 9*, 1318–1324.

Klingspor, I., 1976: Radiometric age-determination of basalts, dolerites and related syenite in Skåne, southern Sweden. *Geologiska Föreningens i Stockholm Förhandlingar 98*, 195–215.

Kornfält, K.-A., Bergström, J., Carserud, L., Henkel, H., & Sundquist, B., 1978: Beskrivning till berggrundskartan och flygmagnetiska kartan Kristianstad SO. *Sveriges geologiska undersökning Af 121*, 1–120.

Lindh, A., Solyom, Z. & Johansson, I., in press: The question of chemical homogenity among basic hypabyssals along the Scandinavian 'Grenville–Gothian Front'. *Sveriges geologiska undersökning C.*

McWilliams, M. O. & Dunlop, D. J., 1978: Grenville paleomagnetism and tectonics. *Canadian Journal of Earth Sciences 15*, 687–695.

Morgan, G. E., 1976: Palaeomagnetism of a slowly cooled plutonic terrain in western Greenland. *Nature 259*, 382–385.

Mulder, F. G., 1971: Paleomagnetic research in some parts of central and southern Sweden. *Sveriges geologiska undersökning C 653*, 1–24.

Murthy, G. S. & Deutsch, E. R., 1975: A new Precambrian paleomagnetic pole for northern Europe. *Physics of the Earth and Planetary Interiors 11*, 91–96.

Norin, R., 1934: Zur Geologie der südschwedischen Basalte. *Meddelanden från Lunds Universitets Geologiska-Mineralogiska Institution 57*, 1–174.

Patchett, P. J., 1978: Rb–Sr ages of Precambrian dolerites and syenites in southern and central Sweden. *Sveriges geologiska undersökning C 747*, 1–63.

Patchett, P. J. & Bylund, G., 1977: Age of Grenville belt magnetisation: Rb–Sr and palaeomagnetic evidence from Swedish dolerites. *Earth and Planetary Science Letters 35*, 92–104.

Pesonen, L. J. 1979: Paleomagnetism of late Precambrian Keweenawan igneous and baked contact rocks from Thunder Bay district, northern Lake Superior. *Bulletin of the Geological Society of Finland 51*, 27–44.

Piper, J. D. A., 1979: A palaeomagnetic survey of the Jotnian dolerites of central-east Sweden. *Geophysical Journal of the Royal Astronomical Society 56*, 461–471.

Piper, J. D. A., 1980: Analogous upper Proterozoic apparent polar wander loops. *Nature 283*, 845–847.

Poorter, R. P. E., 1972: Palaeomagnetism of the Rogaland Precambrian (southwestern Norway). *Physics of the Earth and Planetary Interiors 5*, 167–176.

Poorter, R. P. E., 1975: Palaeomagnetism of Precambrian rocks from southeast Norway and south Sweden. *Physics of the Earth and Planetary Interiors 10*, 74–87.

Pullaiah, G., Irving, E., Buchan, K. L. & Dunlop, D. J., 1975: Magnetization changes caused by burial and uplift. *Earth and Planetary Science Letters 28*, 133–143.

Quensel, P. 1960: Vaggerydssyeniten. *Sveriges geologiska undersökning C 576*, 1–38.

Schweitzer, C., 1975: *Vergleich mehrerer Methoden zur Bestimmung der Intensität des Erdmagnetfeldes an rezenten Laven und ihre Anwendung auf mesozoische und paläozoische magmatische Gesteine.* 157 pp. Ludwig-Maximilians-Universität zu Mün-

chen. [Unpublished Ph. D. thesis.]
Spall, H. 1973: Review of Precambrian palaeomagnetic data for Europe. *Earth and Science Planetary Letters 18*, 1–18.
Springer, N., 1979: En geokronologisk og geokemisk undersögelse af Karlshamnsgraniten, Sverige. *Dansk Geologisk Forening, Årsskrift for 1979*, 78–83.
Stanfors, R., Bergström, S. M., Holmberg, S., Lindström, M. & Hageltorn, B., 1966: *Karta över Skånes berggrund*. Geologiska institutionen, Lund.
Steiger, R. H. & Jäger, E., 1977: Subcommision on Geochronology: Convention on the use of decay constants in geo- and cosmochronology. *Earth and Planetary Science Letters 36*, 359–362.
Watkins, N. D., & Haggerty, S. E., 1967: Primary oxidation variation and petrogenesis in a single lava. *Contributions in Mineralogy and Petrology 15*, 251–271.
Welin, E. & Blomqvist, G., 1966: Further age measurements on radioactive minerals from Sweden. *Geologiska Föreningens i Stockholm Förhandlingar 88*, 3–18.
Welin, E. & Lundqvist, T., 1975: K–Ar ages of Jotnian dolerites in Västernorrland County, central Sweden. *Geologiska Föreningens i Stockholm Förhandlingar 97*, 83–88.
Wilson, R. L., 1972: Palaeomagnetic differences between normal and reversed field sources, and the problem of far-sided and right-handed pole positions. *Geophysical Journal of the Royal Astronomical Society 28*, 295–304.
Zeck, H. P. & Malling, S., 1976: A major global suture in the Precambrian basement of SW. Sweden. *Tectonophysics 31*, T 35–40.